MOHAMMED VI
FACULTY OF MEDICINE
UM6SS - CASABLANCA

ADRENAL SEMIOLOGY

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Semiology: Endocrinology Module

1st Semester
Academic year: 2026-2027

MOHAMMED VI UNIVERSITY
OF SCIENCES AND HEALTH
UM6SS

ENDOCRINE SEMIOLOGY

1- Investigations of the adrenal gland
2-Cushing syndrome
3-Adrenal insufficiency
4- Hyperaldosteronism (adenoma)
5-Pheochromocytoma

Objectives

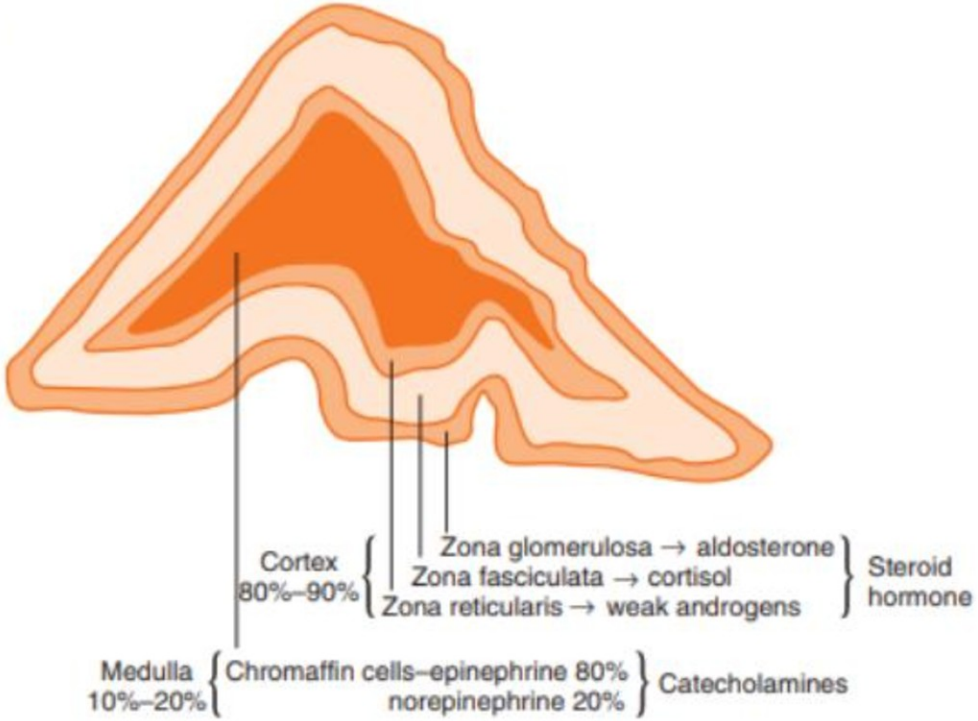
- Explain the clinical and biological manifestations of the main adrenal disorders

- List the main etiologies of these different disorders

I- Exploration of the adrenal gland

REMINDER

2- Histology of the adrenal glands



Zonation and corresponding endocrine function of the adrenal gland.

REMINDER

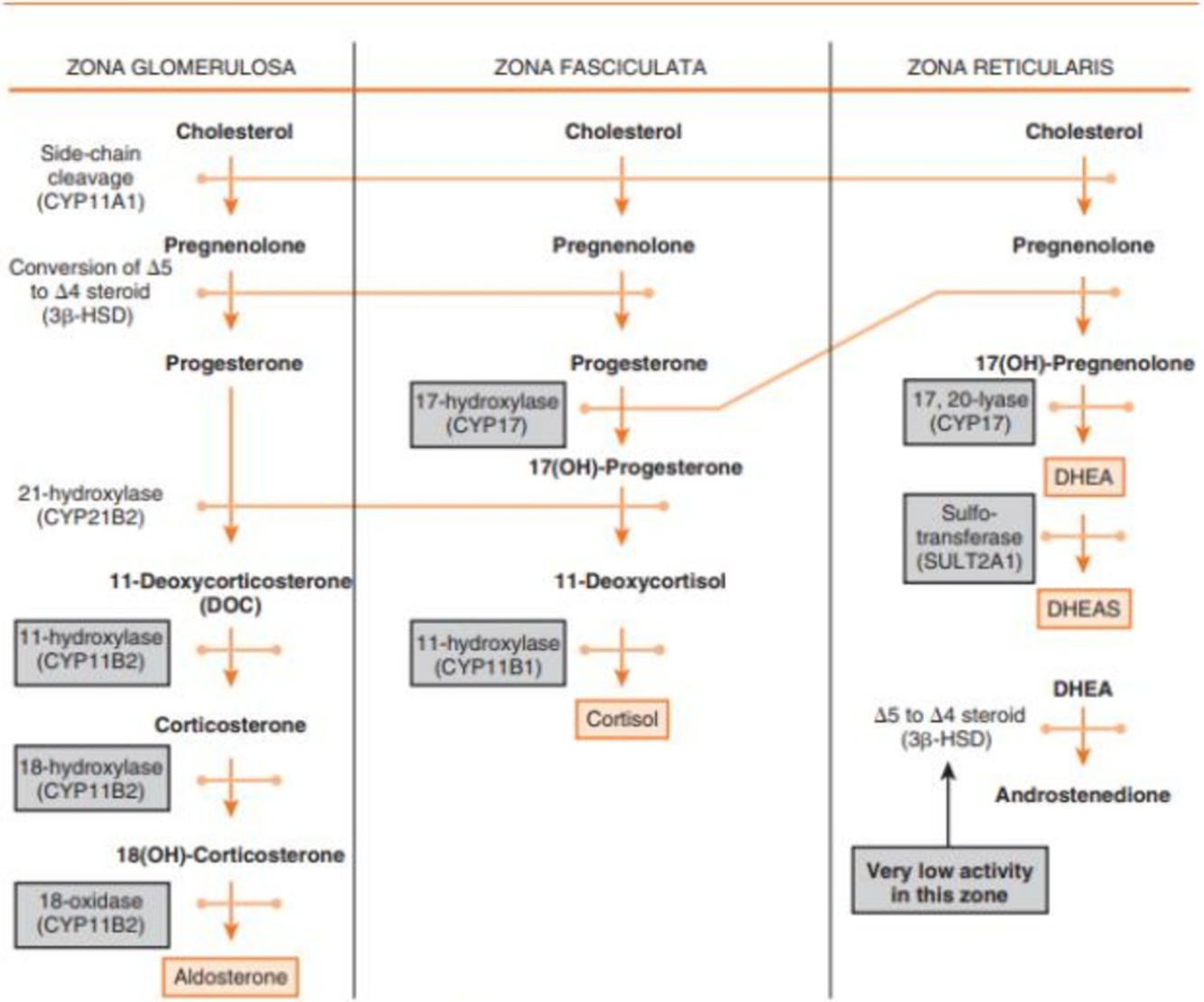
3- Physiology: *Steroidogenesis*

3 types of secretions produced by the adrenal cortex:

- Glucocorticoids: Cortisol and deoxycortisol
- Mineralocorticoids: aldosterone, corticosterone
- Androgens

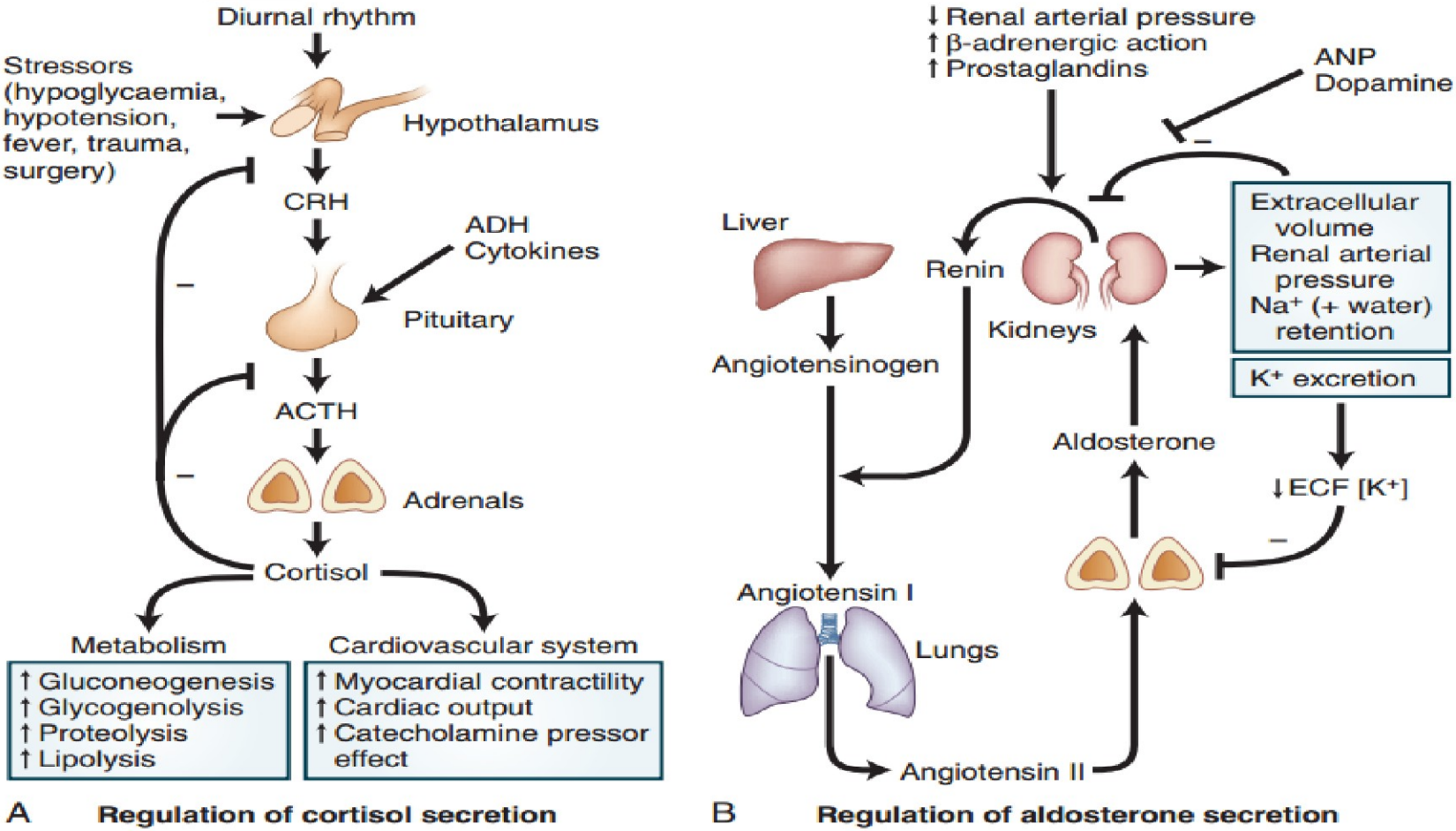
REMINDER

3- Physiology: *Steroidogenesis*



REMINDER

3- Physiology: *Regulation of adrenal hormone secretion*



REMINDER

3- Physiology: *Physiological effects of corticosteroids*

A- Glucocorticoids

- Metabolic actions: carbohydrate, protein, lipid, hydro-electrolytic metabolism

- Neurologic and muscular effects, growth.

B- Adrenal androgens

Minor role, compared to gonadal sex hormones

C-Mineralocorticoid

water and electrolytes metabolism +++

-Na+ reabsorption and K+ excretion

-Acid-base balance

Investigations of the adrenal gland

1- Clinical examination :

In the physiological state, the adrenals are not accessible to clinical examination

Unless the tumor is so large

Search for clinical signs of hormonal hypo or hypersecretion

Investigations of the adrenal gland

2- Investigations:

1- Adrenal CT scan /MRI:

Examination of choice for adrenal exploration:

-Visualization of normal adrenals (inverted y-shape)

-Pathological situations:

-Tumors: characterization of the tumors suggests their natures

-Hyperplasia: homogeneous increase in volume with preserved morphology

-Adrenal atrophy

Investigations of the adrenal gland

2- Investigations:

2- Adrenal scintigraphy

Injection of radioactive marker, with adrenal gland uptake

- **Normal fixation**
- **Hyperplasia: increased volume with bilateral or unilateral homogeneous fixation**
- **Tumor or metastasis**

Investigations of the adrenal gland

2- Investigations:

3- ultrasound

Bad resolution
Does not visualize normal adrenals
Detect tumors >5 cm

4- X-ray of the urinary tree

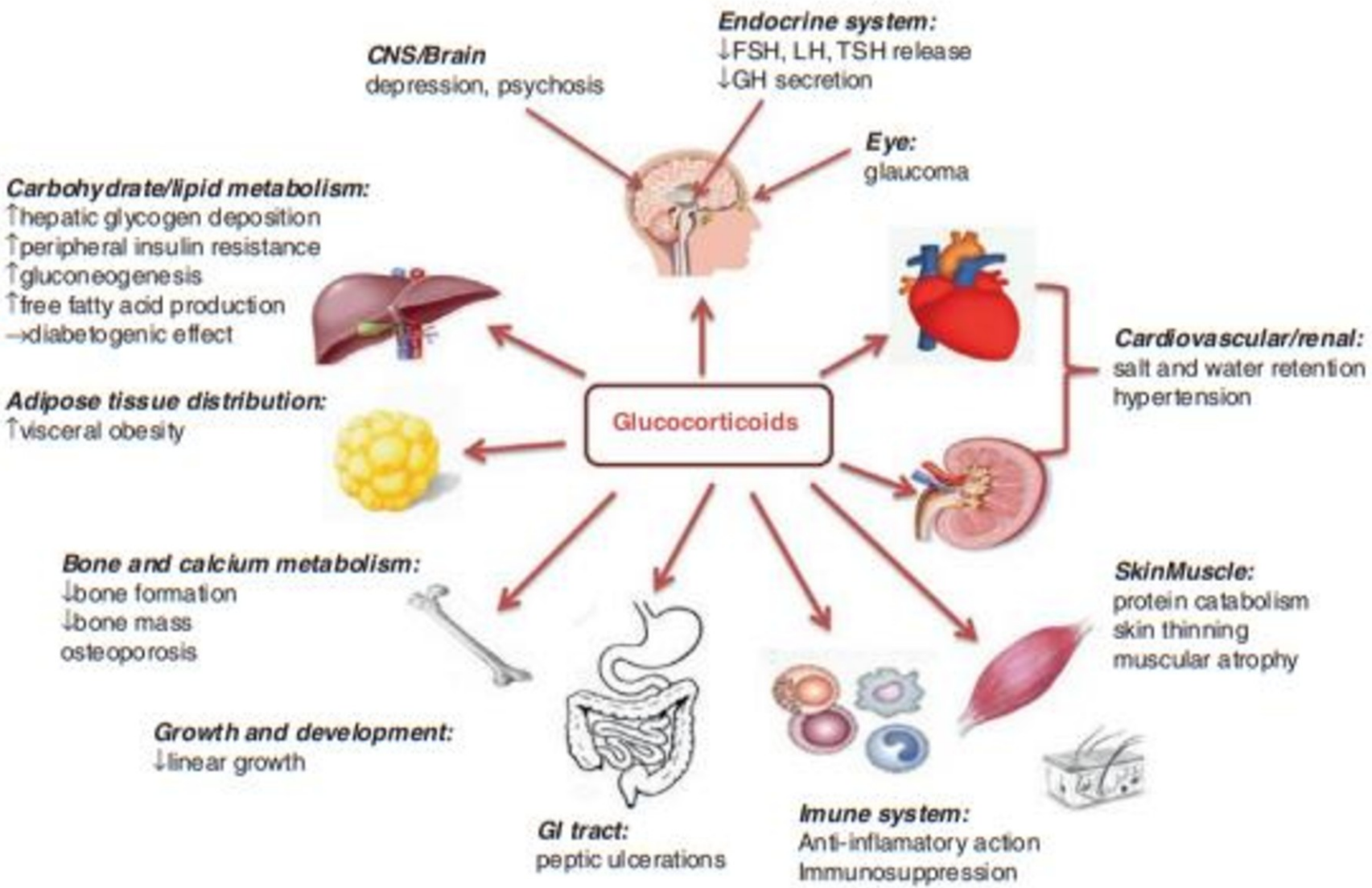
Adrenal area studies (calcifications)

HYPERCORTISOLISM

DEFINITION

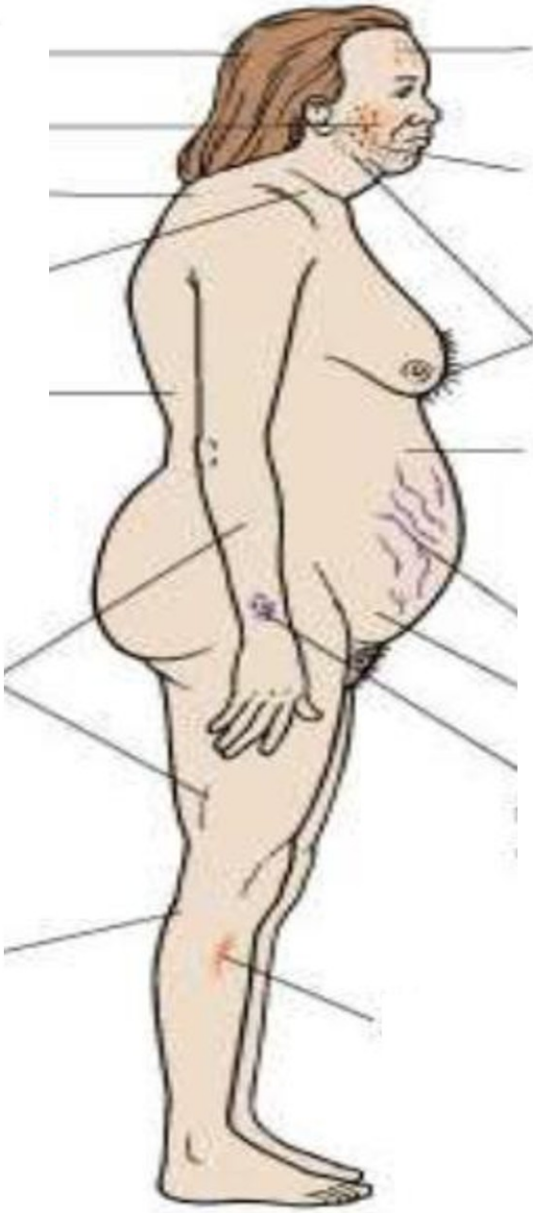
- all the clinical and biological manifestations caused by a chronic excess of glucocorticoids.
- Adrenal hypercortisolism is called Cushing's Syndrome

PATHOPHYSIOLOGY



Clinical manifestations

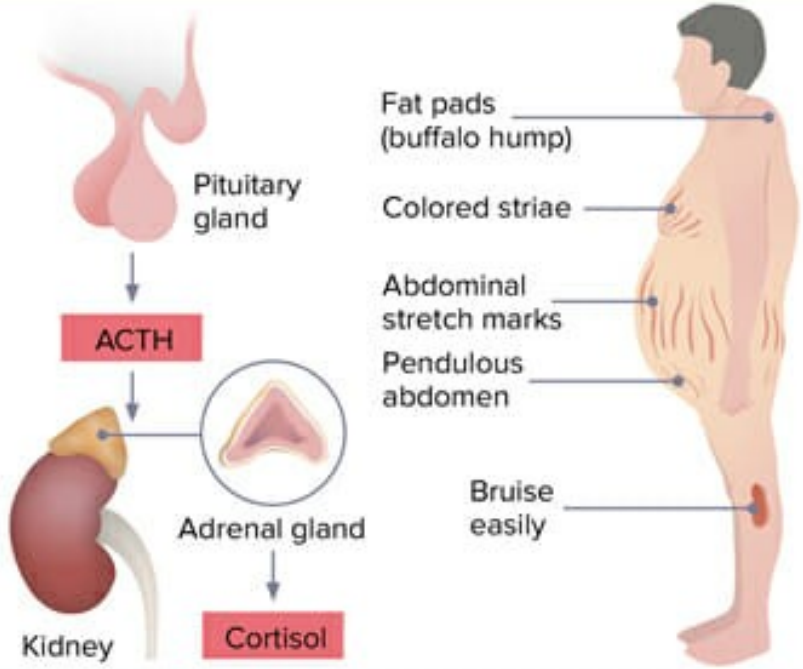
- **1- Clinical features**



CUSHING DISEASE AND CUSHING SYNDROME

Symptoms
(same for Cushing syndrome and Cushing disease)

1. Early: diabetes, high blood pressure, depression
2. Thin extremities from bone and muscle wasting
3. Round face
4. Fat hump between shoulders
5. Weak muscles, skin and bone thinning
6. Increased fat around neck
7. Large pendulous abdomen, obesity
8. Colored/purple striae on abdomen, breast, hips, and axillae
9. Easily bruising
10. Decreased libido, changes in menstrual cycle
11. Hirsutism

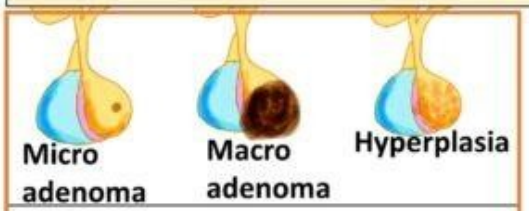


Comparison

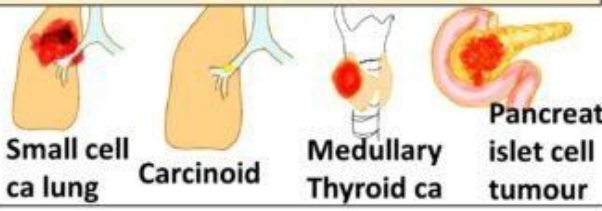
Cushing disease	Cushing syndrome
Causes of Cushing disease	**Causes of Cushing syndrome**
• Pituitary tumor • Adrenal tumor • Ectopic ACTH-producing tumor (*ACTH: adenocorticotropic*	• Long-term use of glucocorticoid medications • Exposure to high levels of cortisol

Causes of CUSHING SYNDROME

ENDOGENOUS



Pituitary: 60-70%



Ectopic hormones: 5-10%

@VijayPatho

EXOGENOUS

Administration of exogenous glucocorticoids

Treatment of chronic immune and inflammatory disorders

CUSHING'S DISEASE → **Increased secretion of ACTH** → **Stimulates adrenal gland** → Adrenal Hyperplasia

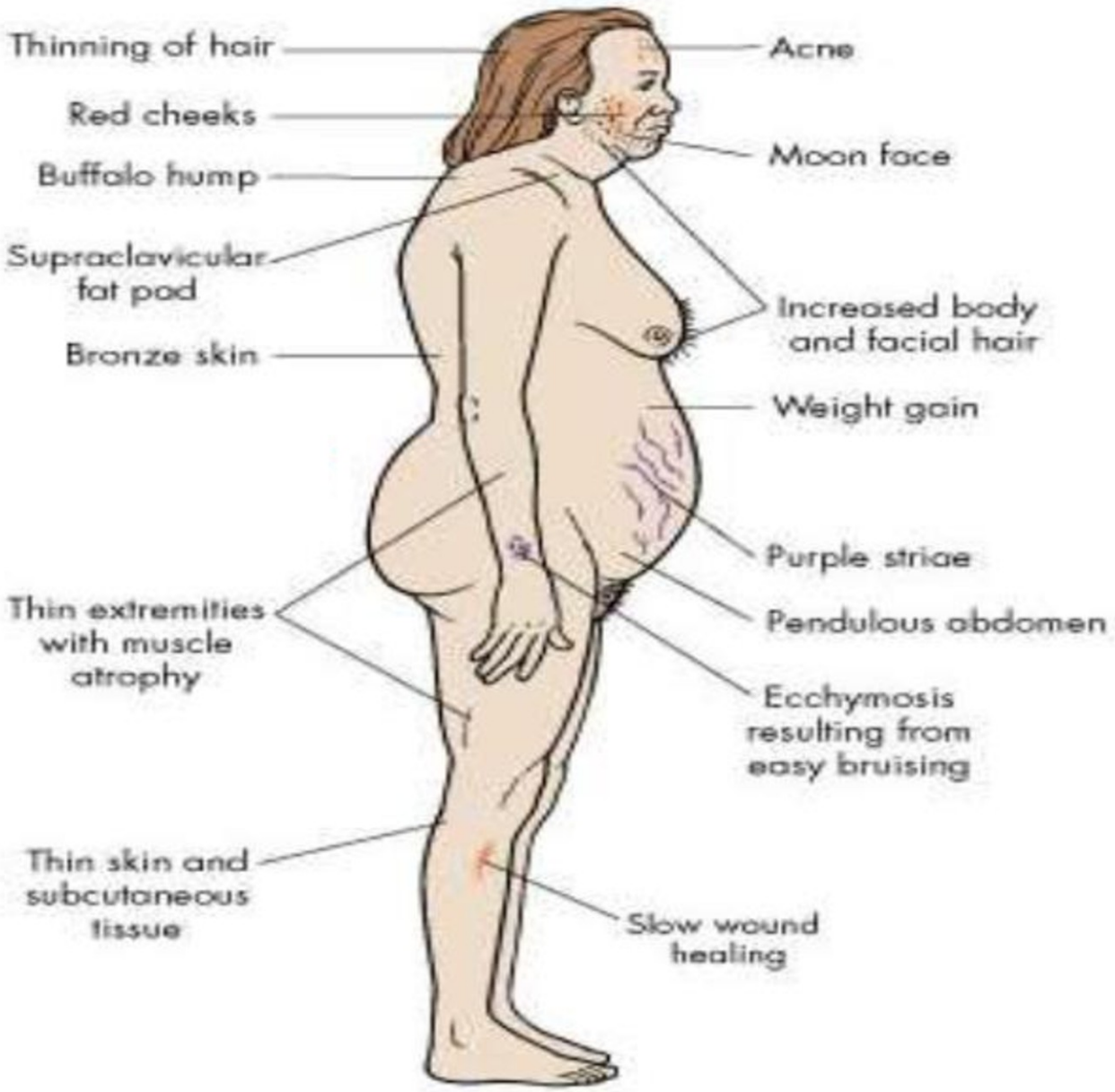
ACTH independent → Adrenal adenoma (10-20%), Adrenal carcinoma (5-7%)

ACTH independent Adrenal Hyperplasia Familial / Sporadic VERY RARE → Adrenal Hyperfunction (hypercortisolism)

→ **CUSHING'S SYNDROME**

Clinical manifestations

- **1- Clinical features**



Clinical manifestations

- **1- Clinical features**

Disorders of fat distribution

- variable weight gain, often moderate
- Central obesity (rounded face, adiposity of the trunk)
- buffalo hump
-absence of excessive fat in the arms and legs

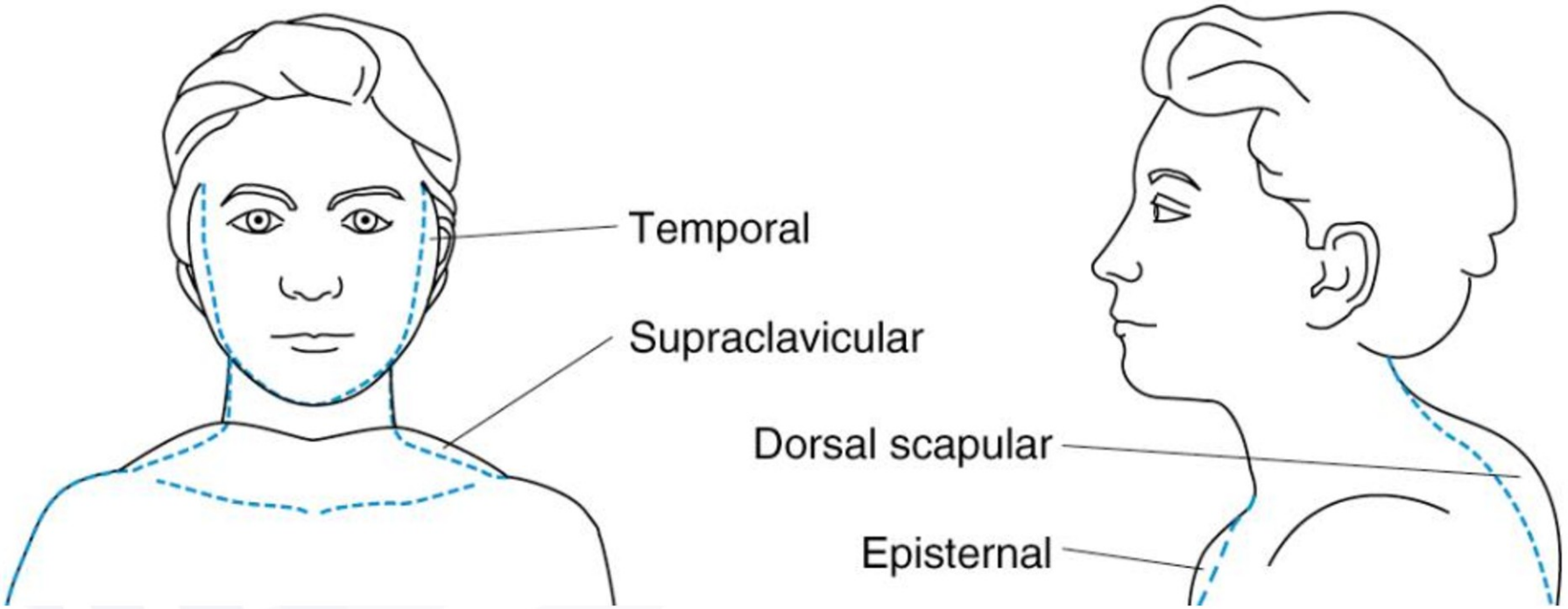
Muscle wasting

- often proximal and is usually most prominent in the lower extremities (myopathy). Hypercortisolism is associated with both low fat-free muscle mass and low total body protein

Clinical manifestations

- **1- Clinical features**

Particular fat deposits



Clinical manifestations

- **1- Clinical features**

Skin atrophy

- fine, fragile skin
- stretch marks, pink or purple, large (> 1 cm), on abdomen, axillar area, breasts
- ecchymoses (due to capillary fragility) bruising occurred without obvious injury
- Facial plethora

Hypertension

moderate or severe, resistant to therapeutics

Psychiatric repercussions

- euphoria with sometimes delirium, mania
- depressive syndrome with suicidal thoughts
- emotional lability, paranoia

Clinical manifestations

- **1- Clinical features**

Bone catabolism

- pain, especially spinal pain
- osteoporosis with vertebral compression
-Pathologic fractures

Reproductive Dysfunction

- folliculitis, acne, seborrhea
- hirsutism
- alopecia
- In women: Menstrual irregularities: oligomenorrhea, amenorrhea
-In men: ↓ libido, impotence ± some degree of gynecomastia;

Vascular and hematological disorders

- Facial plethora(vasodilatation)

Clinical manifestations

- **1- Clinical features**

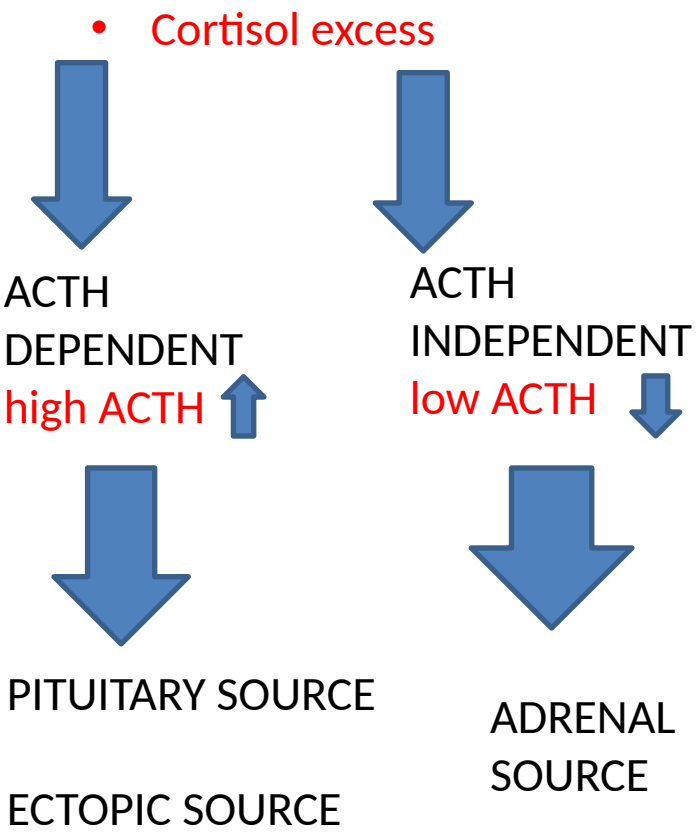
Other Features

- Impaired glucose tolerance/Diabetes Mellitus (30%)
- Hyperlipidemia
- ↑ susceptibility to infections

What we observe: clinic

- Weight excess
- Troncular obesity
- Moon face
- Purple striae
- Hypertension
- Diabetes or pre-diabetes
- Acnea
- Muscle atrophia (limbs)!
- !! Psychiatric problems: depression- suicide risk or euphoria/delirium
- **High risk of trombosis**
- **Osteoporosis**
- Hirsutism and menstrual cycle troubles-in women
- Erectile disfunction -men

Diagnosis hypothesis?



Clinical manifestations

- **1- Clinical features**



Clinical manifestations



Clinical manifestations



INVESTIGATIONS

A- Confirmation tests :

two steps: • Initial testing = confirming the presence of Cushing syndrome. • Testing for the cause of Cushing's syndrome.

Initital tests :

1- **Evidence of excessive cortisol secretion:**

24h urinary free cortisol

Midnight Salivary cortisol

2-Disruption of the circadian rhythm of cortisol secretion

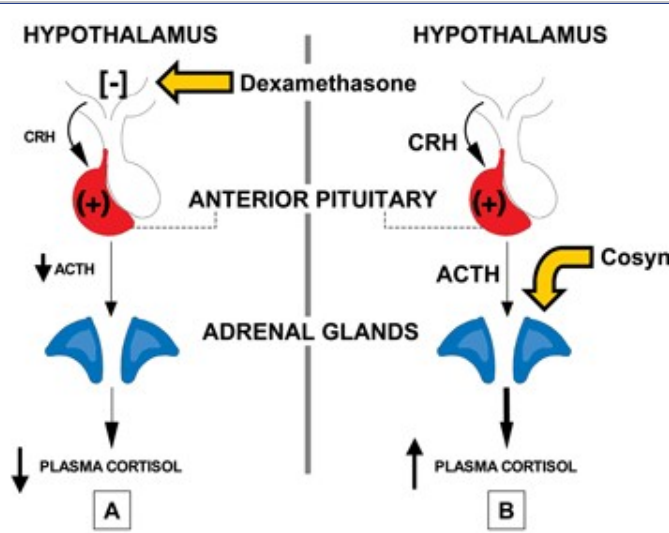
Serum Cortisol cycle (high midnight cortisol)

3-Loss of exogenous corticosteroid feedback on adrenal secretion

✓ 1-mg overnight DXM Suppressive test = the best screening test Procedure

✓ low-dose DST (2 mg/day for 48 h) (LDDST) (the optimal test for those with psychiatric problems, morbid obesity, alcoholism)

INVESTIGATIONS



A- Confirmation tests :

Testing for the cause :

- **Plasma ACTH 8–9 a.m**

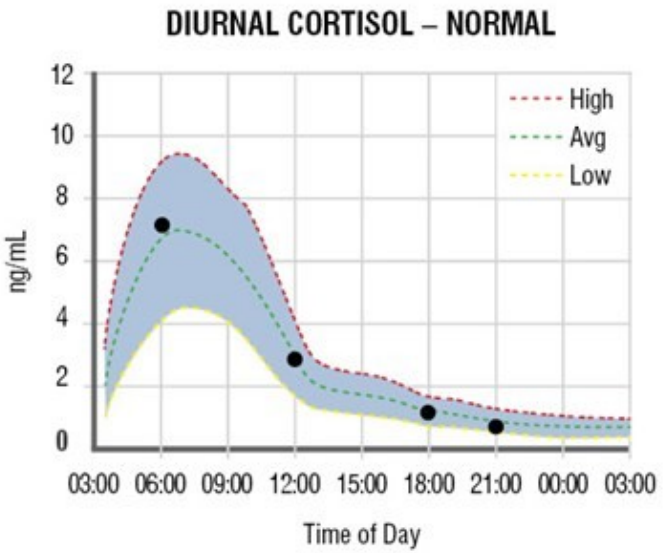
ACTH ↓→ **ACTH-independent Cushing's syndrome** (primary glucocorticoid-secreting adrenal tumors) → the next step = adrenal localizing studies.

• ACTH ↑ or inappropriately normal ⇒ **ACTH secreting neoplasm** [Cushing disease (pituitary ACTH hypersecretion) or ectopic ACTH syndrome]

→ High-Dose DXM Suppression Test (HDDST)

INVESTIGATIONS

Normal Cortisol diurnal profile



B- Others lab tests:

Due to excessive cortisol

- Impaired glucose tolerance or diabetes
- Polycythemia, hyperleukocytosis with PNN, lymphopenia
- Hypokalemia
- hypertriglyceridemia

Etiologies

1- Cushing's disease ACTH independent : Adrenal origin

-Low, undetectable ACTH

- adenomas, carcinomas, adrenal hyperplasia

2- ACTH-dependent cushing's syndrome:

-Normal or elevated ACTH

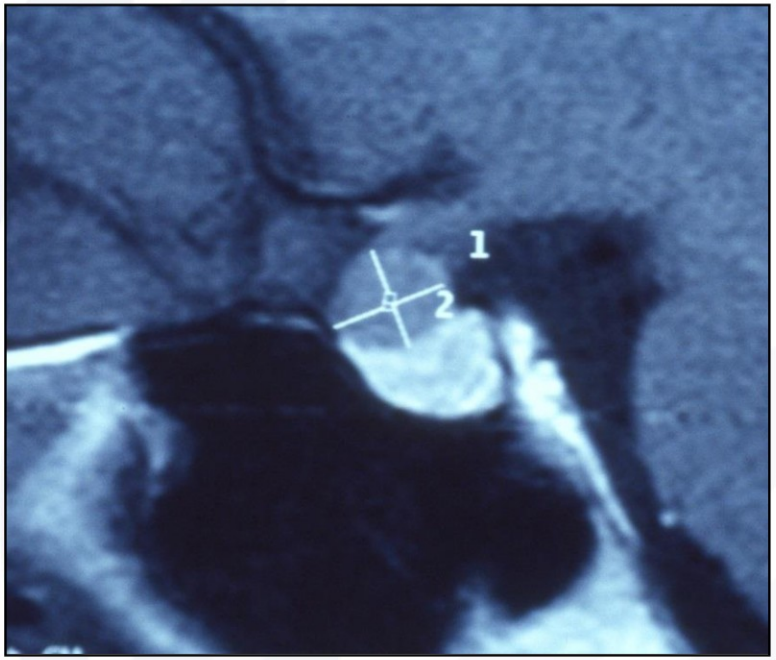
- 2 origins:

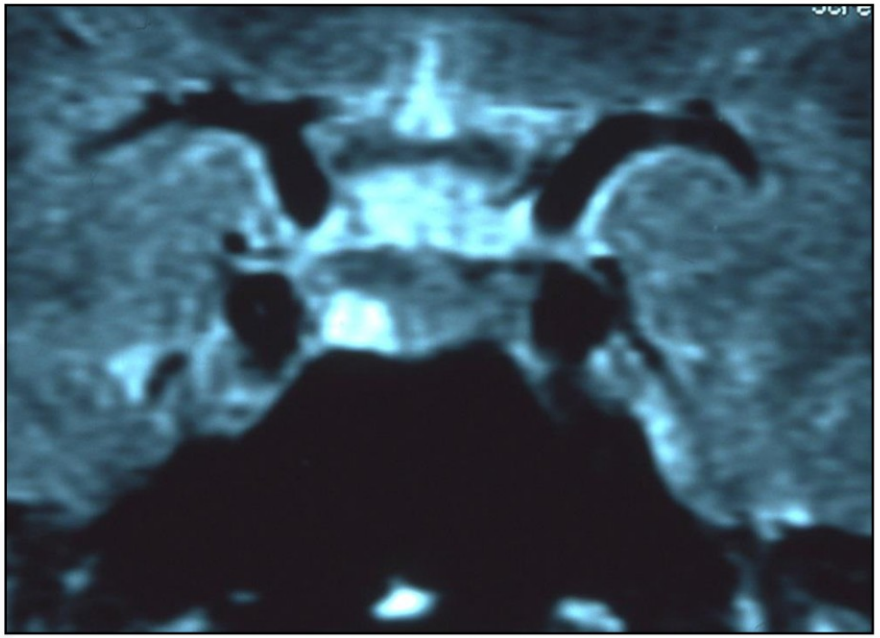
➢ **Pituitary origin:**

Pituitary adenoma (corticotropic cells secreting ACTH)

➢ **ectopic or paraneoplastic cushing ACTH or CRH secretion**

Small-cell lung carcinoma, neuroendocrine tumors


1
2







Complications

- **Metabolic complications:**
  - Impaired glucose tolerance, diabetes
  - Dyslipidemia: high TC, high TG
  - hyperuricemia
- **Cardiovascular complications**
- HTN and its complications (LVH, hypertensive retinopathy)
- **Increased susceptibility to infection:**

  bacterial, viral, mycotic
- **Thromboembolic complications**

Complications

Bone and muscle complications:

-osteoporosis, pathological fractures, vertebral compression

-Aseptic osteonecrosis

-growth retardation in children,

-myopathy

Psyachiatric Complications:

insomnia, agitation, euphoria, maniac or depressive states

ADRENAL INSUFFICIENCY

INTRODUCTION

•Adrenal insufficiency: ***more or less complete deficit of the production and secretion of adrenal hormones***: glucocorticoids, and/or mineralocorticoids, and/or androgens.

•2 origins:
-Primary: lesion affecting both adrenal glands
- Secondary: to an impairment of corticotropic control.

•The clinical picture is often insidious,
•can be revealed in an acute mode, with a life-threatening situation.

•The diagnosis suspected on the basis of clinical and biological elements , is confirmed by the lowering of the 8H cortisol level and if necessary by dynamic tests

Clinical manifestations

A- clinical presentation

Often insidious symptomatology
Non specific → diagnostic delay

1- General weakness

- **Asthenia:** physical, psychological and sexual

Increased during the day, maximum in the evening and during efforts

- **Weight loss:** constant and variable from 2 to 15 kg on average
- **Anorexia**

→invariable features!

2- Gastrointestinal disturbances

- Nausea
- Abdominal pain
- Transit disorders: diarrhea

Clinical manifestations

B- FUNCTIONAL SIGNS

3-Hypoglycemic manifestations

cortisol deficiency

In fasting state, or after prolonged physical effort or in case of intercurrent disease

4- Psychiatric disorders:

Depressive state or even psychotic disorders which may be at the forefront

5- Amenorrhea

Clinical manifestations

C-clinical examination :

1. hyperpigmentation

•Fundamental element of the diagnosis

•Observed in the non-naturally pigmented patient

•Predominantly in sun-exposed areas, and accentuated over pressure areas such as the knuckles, toes, elbows, and knees.

• hyperpigmentation of the buccal mucosa and gums
•pigmentation of the palmar creases, nail beds, nipples, areolae, and perivaginal and perianal mucosa.
• Scars that have formed after the onset of ACTH excess become hyperpigmented, whereas older ones do not.

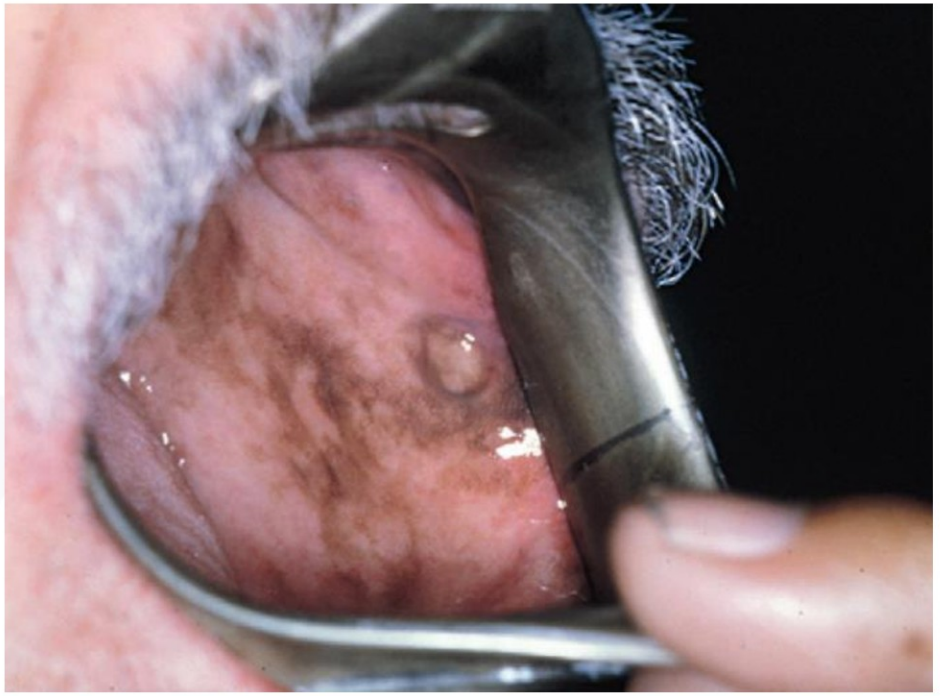














Clinical manifestations

2- Systolic arterial hypotension

- present in about 90% of patients, accompanied by orthostatic symptoms: dizziness and occasionally syncope.

- Positive Schellong test (a reduction in systolic BP of at least 20 mmHg or diastolic BP of at least 10 mmHg within 3 min of standing).

Paraclinical Diagnosis

1) Confirmation assessment:

The 8 am serum Cortisol level :

if < 5ug/dl (80nmol/l) → AI

if > 18ug/dl → exclude AI

if Intermediate values: ?

Normal adrenal function or latent insufficiency that may be revealed during stress.



Requires the use of dynamic tests.

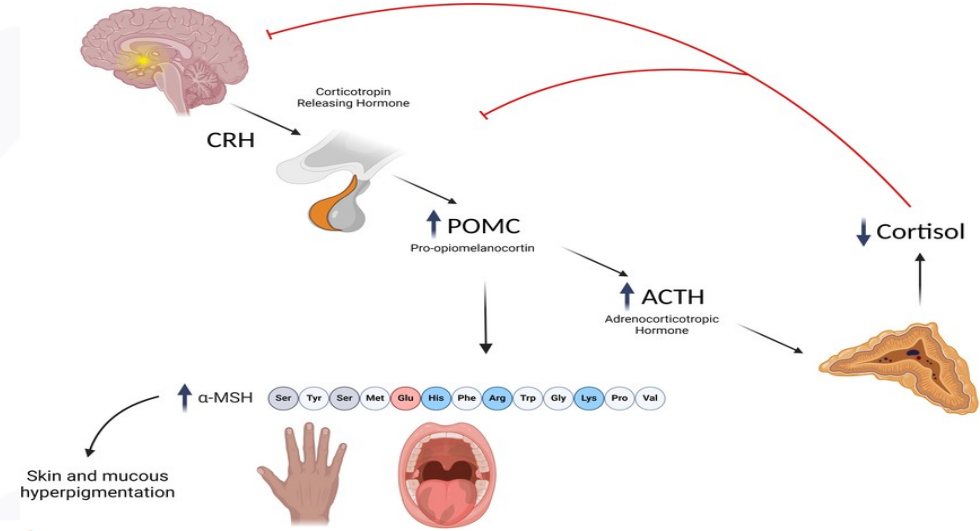
Paraclinical Diagnosis

1) Confirmation assessment:

Standard synacthene tests:

- Cortisolaemia measured at T0 and T60 min after IM injection of 250ug of tetracosactide synacthen 0.25mg

- **Insufficient elevation < 18ug/dl confirms adrenal insufficiency.**



Paraclinical Diagnosis

2) Others tests

Suspicion elements,variable:

- **Electrolyte disorders:**
  - ✓hyponatremia
  - ✓ hyperkalemia.
  - ✓ Hypercalcemia is rare

- **Fasting blood glucose** is usually in the normal low range
- **CBC**: Moderate anemia, normochromic, normocytic.

Paraclinical Diagnosis

3. Etiologic workup:

First step : 8 am ACTH assessment :

high: peripheral origin

normal or low: central origin

Anti 21 hydroxylase self antibody assay

Adrenal CT: Normal or reduced size of the adrenals, calcifications

Urinary tree Xray : calcifications

Etiologies

A- Primary adrenal insufficiancy:

- **Addison's disease:** Anti 21 hydroxylase antibodies
- **Adrenal tuberculosis:** calcifications on imaging
- **Other etiologies:** Hemorrhage, Metastasis, Infection.

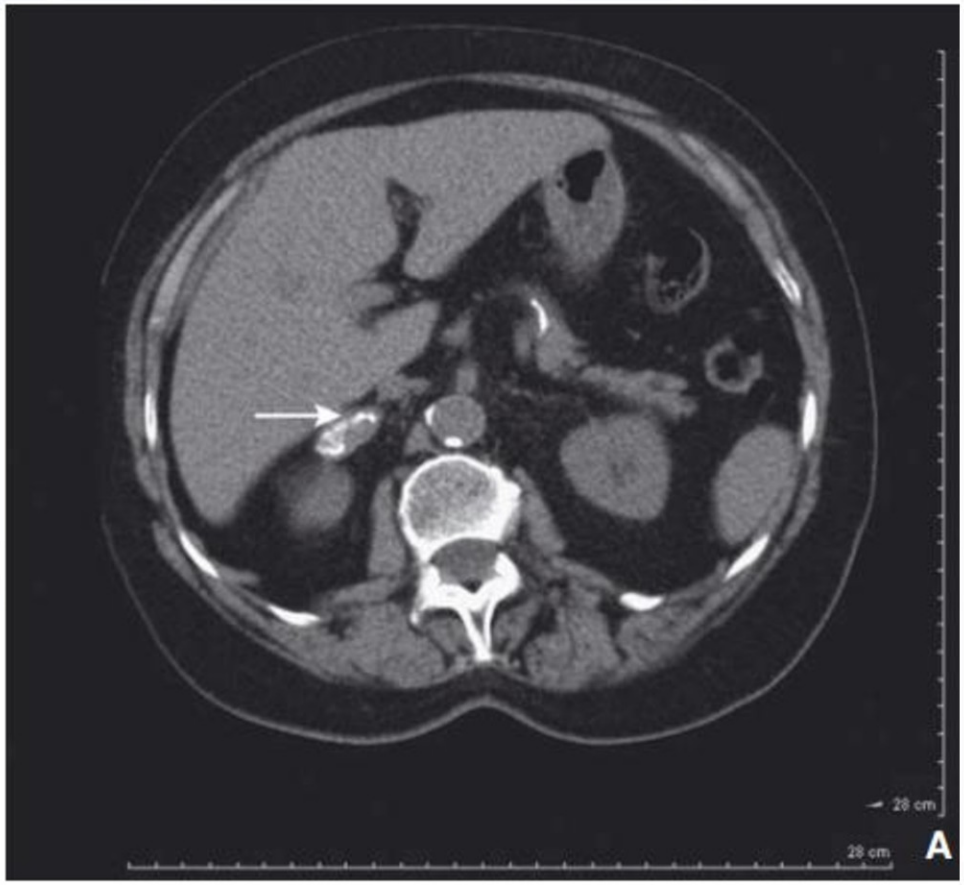
COMPUTER TOMOGRAPHY IN ADRENAL INSUFFICIENCY

Adrenal structure and size
Tumoral etiology



3 *Atteinte auto-immune : surrénale gauche de petite taille (rétraction corticale).*

- **Adrenal tuberculosis:**





CLINICAL FORMS

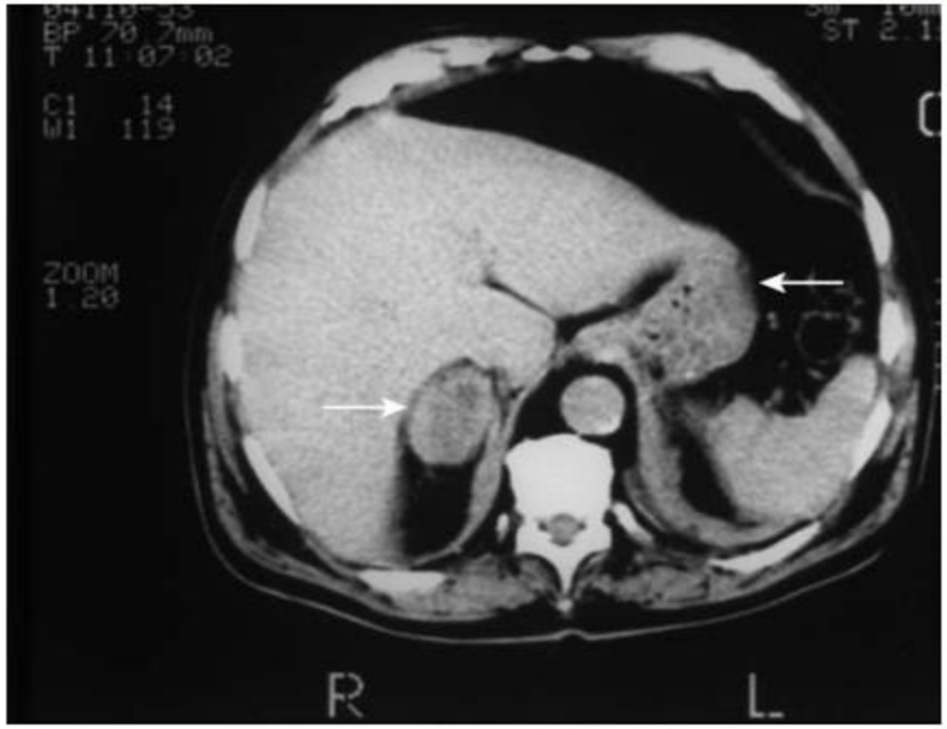


Figure 10. Aspect scanographique d'une hémorragie bilatérale des surrénales. Noter l'aspect volumineux et hétérogène des glandes surrénales.



Figure 11. Aspect scanographique de métastases surrénaliennes bilatérales d'un carcinome colique révélées par un déficit surrénalien primaire (A, B).

Etiologies

B- Secondary adrenal insufficiency

Many of the symptoms are common, but there are fundamental differences:

1) Clinical features :

Absence hyperpigmentation: ACTH deficiency, pale depigmentation particularly evident in brown subjects, in normally pigmented areas

Hypoglycemic events are more frequent

Hypotension and digestive disorders are clinical manifestations rarely observed because of the conservation of mineralocorticoid secretion

CLINICAL FORMS

B- Secondary adrenal insufficiency

2) Laboratory tests:

No biological stigma of volume depletion, nor hyperkalemia nevertheless we have:

hyponatremia
Normocytic normochromic anemia, moderate lymphocytosis

3) ACTH assessment:

•ACTH: low normal with low cortisol levels

COMPLICATIONS

ACUTE ADRENAL INSUFFICIENCY

- This is the natural evolution of slow adrenal insufficiency
- Revealing latente AI on the occasion of an intercurrent disease

- Acute episode: Infection, trauma, surgery

- Therapeutic error:
  - cessation of treatment;
  - taking diuretics or abuse of laxatives;
  - Salt-free diet.

- In case of hemorragic cause on healthy adrenals.

COMPLICATIONS

Clinical picture:

- ***Cardiovascular collapse*:** major arterial hypotension, pallor, sweating, cold extremities, thready pulse.
- ***Intense digestive signs*:** anterior abdominal and lumbar fossa pain, vomiting, profuse diarrhea...
- ***Neurological disorders*:** altered mental state (extreme adynamia, drowsiness, mental confusion or even coma) in severe forms.
- ***Signs of dehydration***.

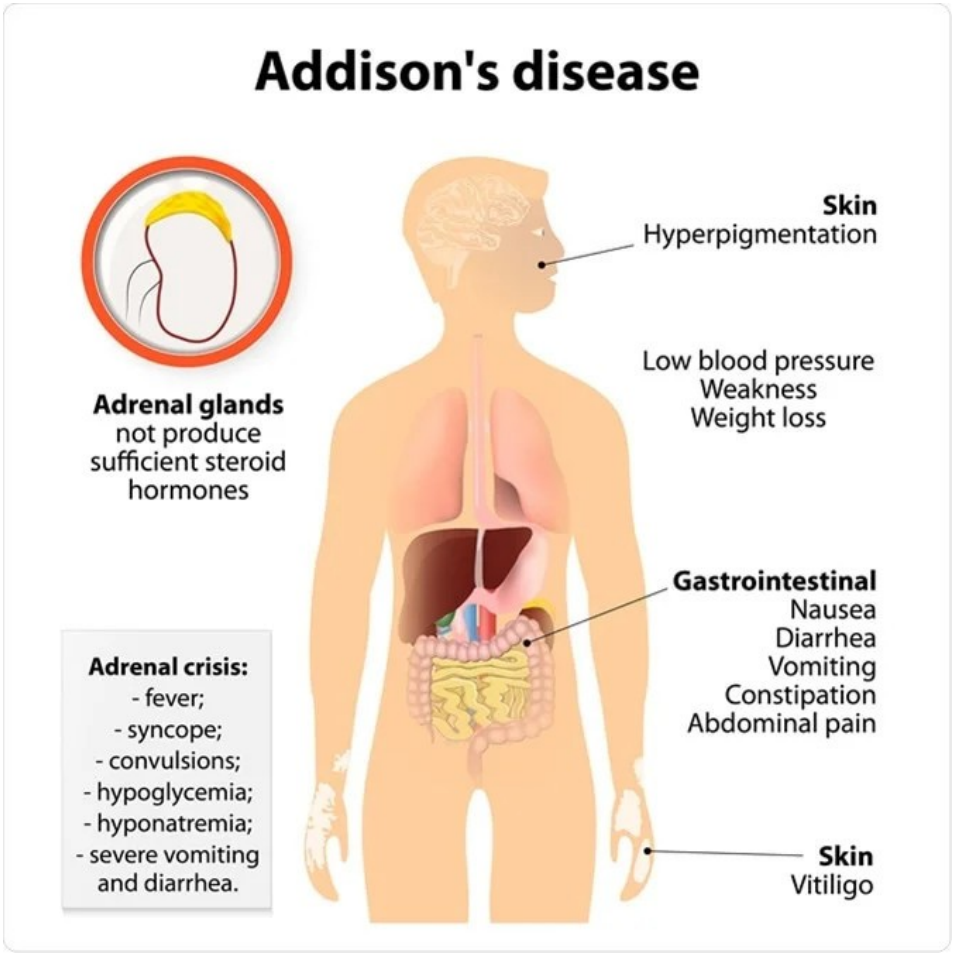
COMPLICATIONS

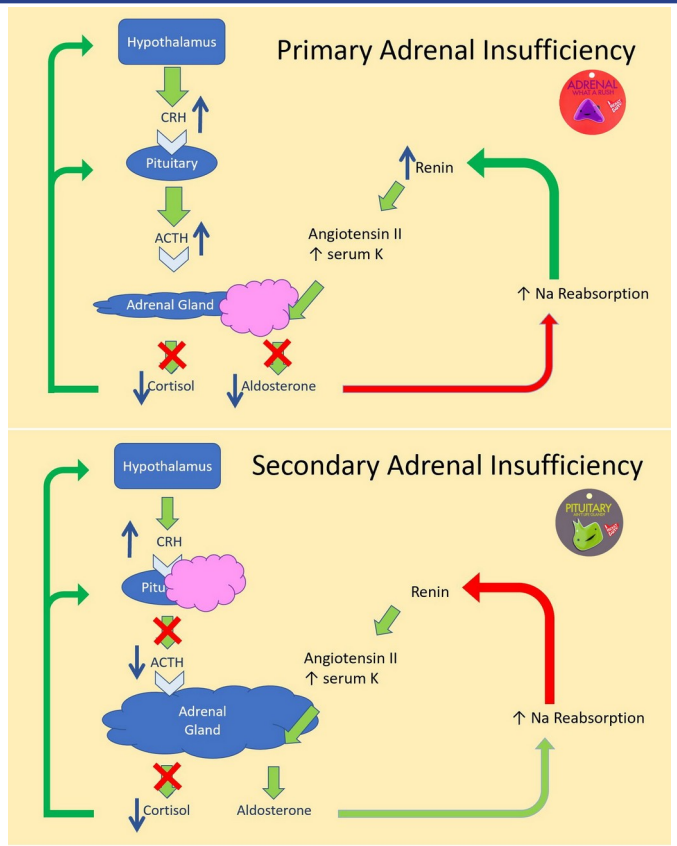
Lab tests:

-Hyponatremia
-hyperkalemia responsible for ECG disorders (QRS prolongation, large T-wave) with a risk of death.
- Signs of hemoconcentration (elevated hematocrit, protidemia and urea-FRI)
-Hypoglycemia.

- The diagnosis is confirmed by the therapeutic response to the administration of high doses of hydrocortisone.
- Hormone assays, performed without delay of treatment.

ADRENAL INSUFFICIENCY



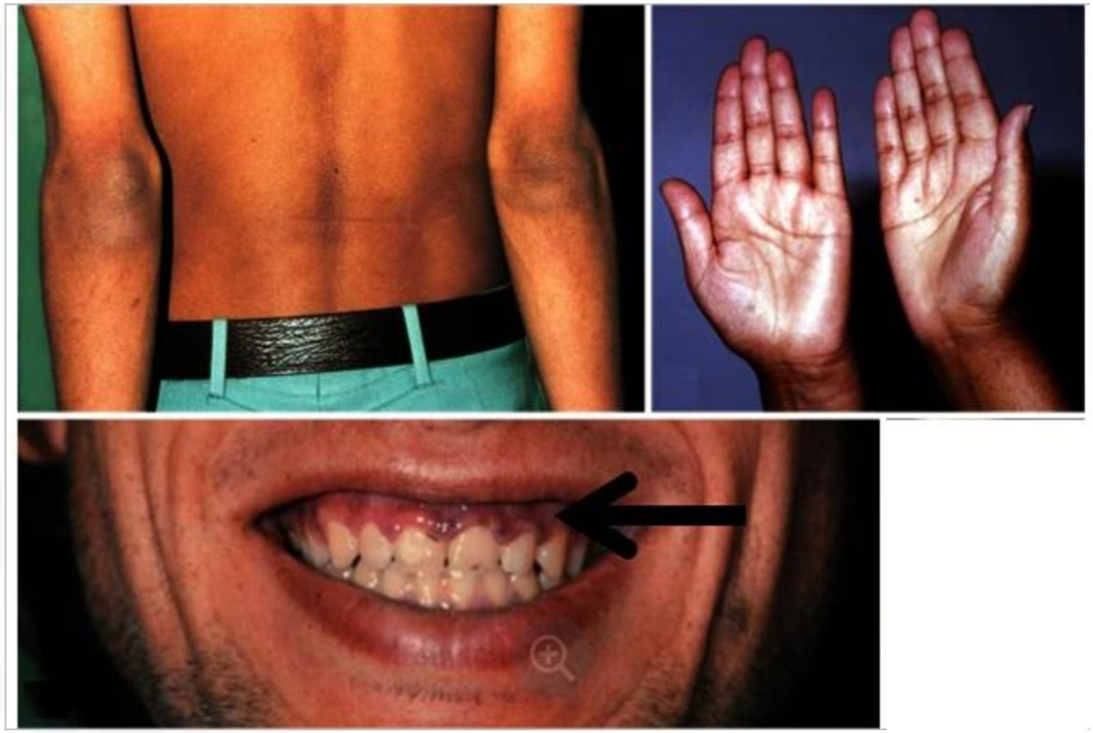


AUTOIMMUNE
HIGH ACTH= HYPERPIGMENTATION
LOW CORTISOL+ LOW ALDOSTERON

Secondary adrenal insufficiency:

- Pituitary cause- low ACTH= no hyperpigmentation
- Low cortisol
- Normal Aldosteron (less severe hypotension and dyselectrolitemia)

Clinical case



- **Clinical Examination:**

  - BMI: 17kg/m2; BP: 100/60mmHg;

  - Gc : 0.65g/l

  - hyperpigmentation: in sun-exposed areas; pressure areas,buccal mucosa and gums and palmar creases.

Clinical case

- Cortisol level at 8:00 am: 3,1 ng/ml (N:30-200 ng/ml)

ACTH=300pg/l (10-60)

? Diagnosis

Primary adrenal insufficiency

ALDOSTERONE-PRODUCING ADENOMA

INTRODUCTION

- Benign adrenal tumor (adenoma) of the adrenal cortex **producing excessive aldosterone**

- **Clinical manifestations: systolodyastolic hypertension associated with hypokalemia**

- Clinical Manifestations:

-**HTN**

-**Manifestations related to hypokalemia: Fatigue and muscle cramps, polyuria and polydispia, paresthesia, pseudo paralysis.**

- **Biological tests :**
- **Hypokalemia (< 3.5 mmol/l) or < 3.9 mmol/**
- **Determination of the aldosterone/renin ratio:**

Conditions:

- **Normo-sodium diet**
- **Discontinuation of anti-HTA drugs interfering with the renin angiotensin system**
- **Fasting morning**

- ***Normal < 64***
- ***In case of adenoma: high Aldosterone, low Renin with an Aldosterone/Renin ratio >64***

Morphological exploration:

- CT scan: hypodense mass 1 to 3 cm
- MRI: ≈ CT scan
- Noriodocholesterol scintigraphy

Adenoma: unilateral hyperfixation

PHEOCHROMOCYTOMA

• Definition: Tumors of the chromaffin cells of the adrenal medulla (=pheochromocytomas) or extra-adrenal medulla (= paragangliomas), secreting catecholamines.

- Curable cause of Hypertension
- Complications, mortality related to hypertension
- Paroxysmal hypertension +++
- Severe and resistant hypertension

SYMPATHETIC NERVOUS SYSTEM – CATECOLAMINES SECRETION

- ADRENALINE/ NORADRENALINE PRODUCING CELLS

- PARAGALIOMA POSSIBLE LOCALISATION

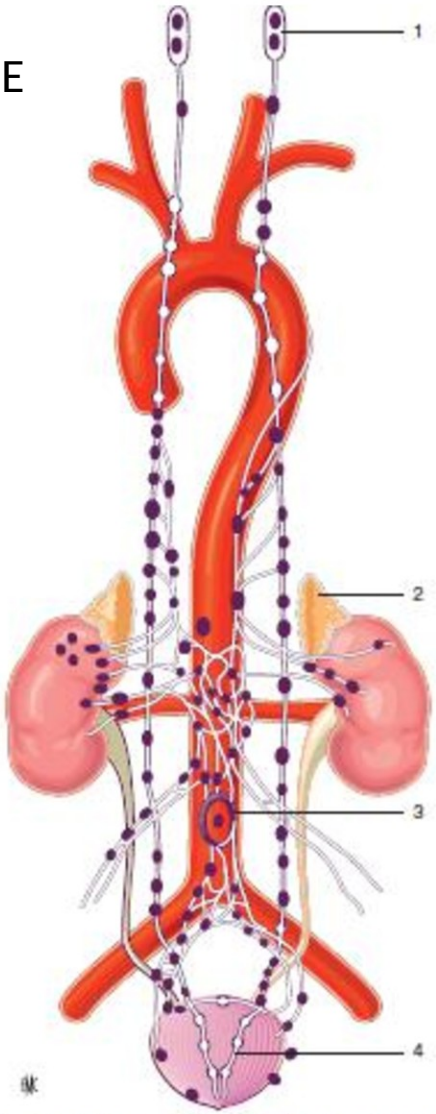


Figure 2. Localisations possibles des phéochromocytomes et des paragangliomes. 1. Paragangliomes tête et cou ; 2. phéochromocytomes (glandes surrénales) ; 3. organe de Zuckerkandl ; 4. paragangliomes vésicaux.

Clinical manifestations

Paroxysmal symptoms+++.

- triggered by tumor compression
- Variable in frequency, duration, intensity

- *headaches, palpitations, pallor, profuse sweating,* cooling of the extremities

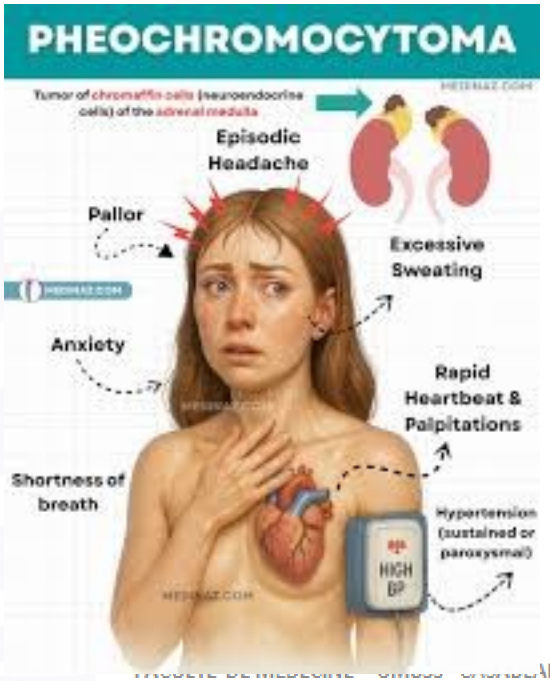
Triad: headache + sweating + palpitations

Clinical manifestations

Paroxysmal symptoms+++.

- anxiety, tremors, sometimes vomiting, disorders visual, abdominal or chest pain

- during these attacks, BP can be considerably higher than usual.



Clinical manifestations

Chronic symptoms

- In case of permanent secretion of the tumor
- with paroxysmal symptoms
- HTANof variable level, resistant to the usual treatments, sometimes associated with *orthostatic hypotension*

.

Signs of hypermetabolism:

- weight loss, fever, thermophobia
- glucose intolerance, even diabetes

Biological check-up

- **In the urine:**

- determination of *catecholamines, especially methoxylated derivatives*

Urinary+++.

-The measurement of VMA is less reliable and is no longer practiced.

- **In the blood:**

-The determination of plasma catecholamines, stress hormones with a short half-life, is difficult to interpret.

- On the other hand, the determination of *metanephrines cad derivatives*

very interesting plasma methoxylates.

Imaging

- CT and MRI examinations of choice, tumor localization
intrabdominal or thoracic in almost all cases and their morphological analysis.

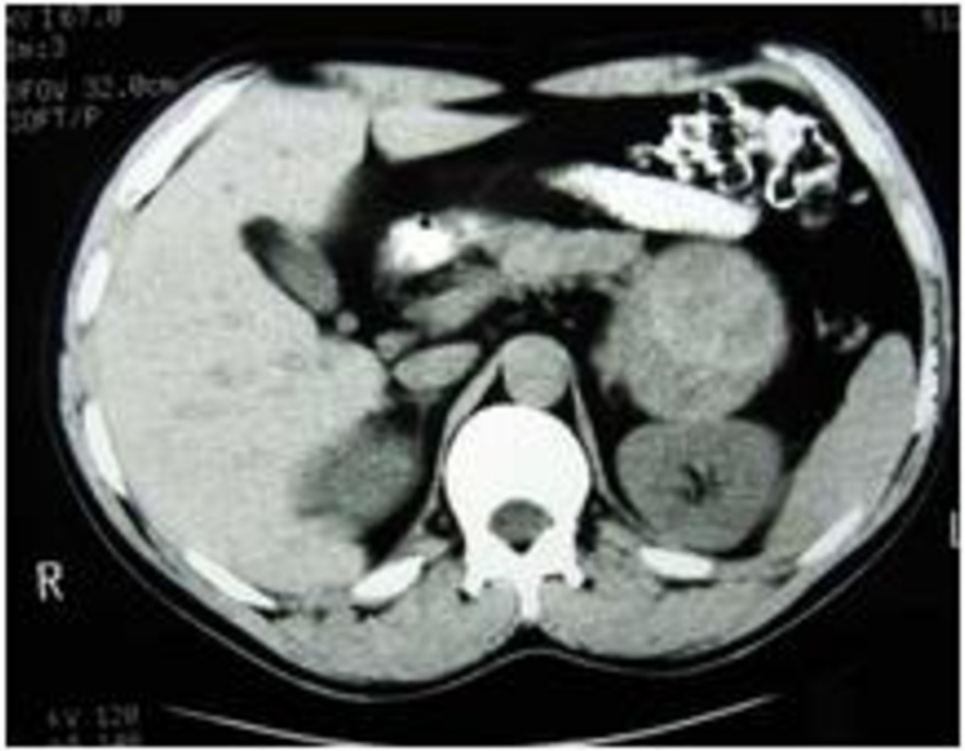
- MIBG (meta iodo benzyl guanidine) scintigraphy
completes the previous techniques: exploration of the whole body in search of multiple locations, ectopic, metastases, reveals very small lesions that may escape the scanner or MRI.

Adrenal computer- tomography

ADENOMA SIZE

ADENOMA STRUCTURE- LIPID

TUMOR EXTENTION



TDM coupe horizontale. Phéochromocytome gauche

MIBG SCINTIGRAPHY- FUNCTIONAL EVALUATION OF THE ADRENAL

- UNILATERAL SECRETION
- ADENOMA/ HYPERPLASIA
- ECTOPIC – NON/ADRENAL SECRETION

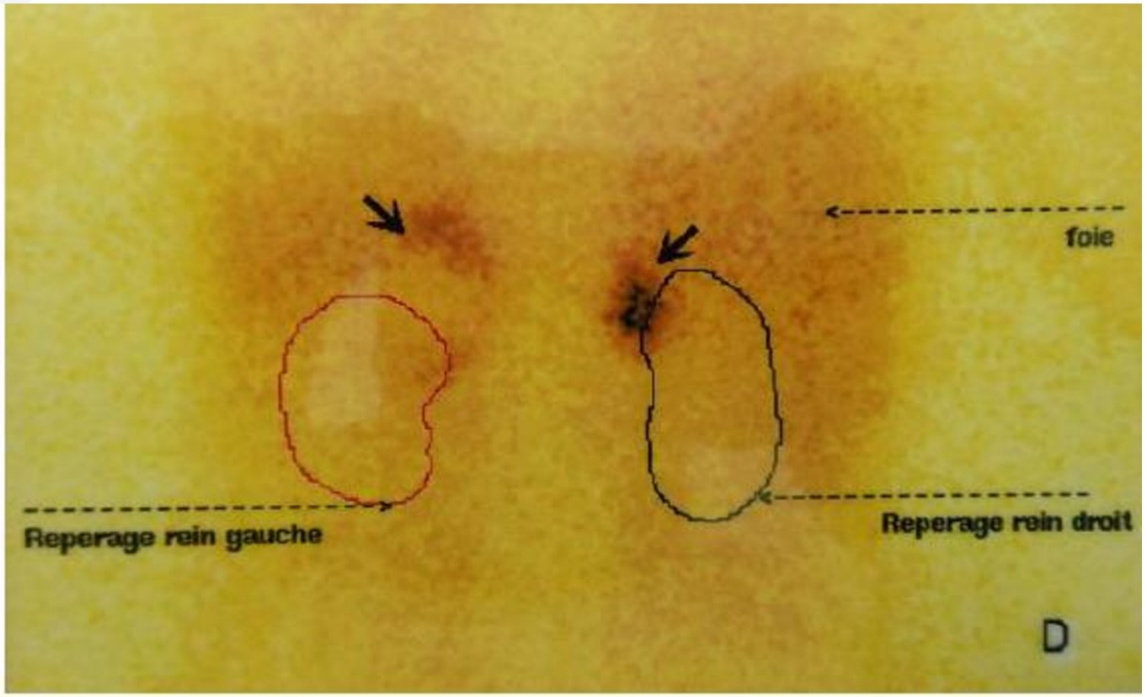
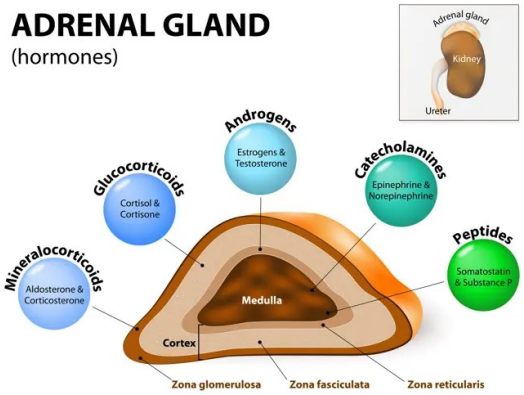


Figure 2: Scintigraphie à la MIBG: Présence d'une hyperfixation surrénalienne bilatérale avec absence de signe scintigraphique en faveur d'une localisation ectopique ou de métastase

SYNTHESIS

- ADRENAL CORTEX HYPERSECRETION: CUSHING, HYPERALDOSTERONISM
- ADRENAL CORTEX INSUFFICIENCY: primary (with hyperpigmentation) or secondary

- HYPERALDOSTERONISM- ALDOSTERON SECRETING ADENOMA



- FEOCROMOCYTOMA